

Thomas Chisica Londoño

Applied Mathematics & Computer Science | Bogotá, Colombia

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RESEARCH INTERESTS

Reinforcement learning, computational neuroscience and collective behaviour. Building simulations and research software to study learning and decision-making through controlled comparisons.

EDUCATION

Universidad del Rosario, Bogotá

2024–present

BSc in Applied Mathematics and Computer Science. Expected graduation: June 2028.

GPA: 4.11/5.00 | Full-tuition merit scholarship.

Relevant coursework: Machine Learning (5.0/5.0), Optimisation, Probability & Statistics, Real Analysis, Linear Algebra, Graph Theory and Differential Equations; Numerical Analysis (in progress).

SELECTED RESEARCH & TRAINING

Reinforcement learning for supply-chain control

2025–present

Undergraduate research with Prof. Alexander Garrido, Universidad del Rosario

- Reconstructed a thirteen-operation supply chain in SimPy; developed Gymnasium environments and PPO experiments to compare control policies under operational disruptions.
- Evaluated 10 PPO checkpoints on 60 held-out streams against a calibration-selected policy with the same action space; identified a comparator mismatch that had overstated the benefit of learning.
- Built automated tests and source-model checks; prepared contributor documentation covering environment interfaces, trajectory export and evaluation procedures to support collaborative experiments.

 [SCRES: simulation and evaluation code](#)

Neuromatch Academy: Computational Neuroscience

July 2026

68-hour interactive course; team project and independent follow-up

- Studied motor-task learning in recurrent neural networks across four connectivity densities and eight paired seeds, using held-out trials with learning disabled for evaluation.
- Implemented an equal-plasticity control, holding the number of trainable recurrent connections fixed across densities. The original density advantage did not recur under this control.
- Coursework: neural encoding/decoding, GLMs, dimensionality reduction, deep learning, signal processing, neuronal dynamics, Bayesian inference, hidden-state models, reinforcement learning and network causality.

 [Motor-RNN experiments and equal-plasticity control](#)

Spectral and geometric analysis of human collective search

2026

Reanalysis of Andrade-Lotero and Goldstone's SODCL experiment and data

- Modelled the 8×8 task board as a spatial graph and compared human search partitions using conductance and information measures, accounting for a degenerate Fiedler eigenspace.
- Reconstructed past-only features for 45 dyads at their fifth target-absent round, removing overlap with late outcomes; checked behavioural measures against source records and added six integrity tests.
- Compared basis-invariant Fiedler and coordinate energies in a model-free audit; found near-redundancy in this sample, leaving predictive value unresolved.

 [Spectral/SODCL: reanalysis and temporal audit](#)

SELECTED RESEARCH & TRAINING (CONTINUED)

Computational analysis of marmot social networks

2026–present

Research with Dr. Adriana Maldonado-Chaparro, Universidad del Rosario

- Built a SQLite data-cleaning pipeline linking social interactions, sightings, residence and relatedness into dyadic tables; examined how observed co-presence represents opportunities for interaction.
- Developed a cross-implementation validation workflow for a legacy NetLogo/R individual-based model using Python and native-runtime checks; identified defects in the implementation of homophily rules.
- Implemented exploratory count and mixed-effects models, group-held-out evaluation and matched-sample comparisons to distinguish grouping effects from sample differences; quantified changes in observation coverage over time.

Collective behaviour: experimental software

2025–present

Research with Prof. Edgar Andrade-Lotero, Universidad del Rosario

- Contributed workflow documentation and smoke-test commands to the PsyNet-based Coordinator and Foragers experiment, supporting installation and reproducible execution of the team’s code.

PROFESSIONAL EXPERIENCE

Lead in Emerging Technologies

June–December 2025

Digital Business Ventures; Witty project with Progresión

- Built a private retrieval-augmented support prototype with FastAPI, FAISS, embeddings and optional LLM rewriting; implemented heuristic routing between retrieved answers and human follow-up.

SELECTED SOFTWARE PROJECTS

HeliOS: educational RISC-V kernel

2025–2026

Contributed to a collaborative C kernel with task scheduling, timer interrupts, a UART shell, synchronisation primitives and heap allocation; includes QEMU/OpenSBI smoke tests.

[!\[\]\(83bbbd261710c59db0214aa27b2edc0d_img.jpg\) Kernel, architecture and test scripts](#)

ContratIA Abierta: public-procurement review tools

2026

Developed a data-processing and review pipeline for Colombian SECOP open data using Polars, PostgreSQL, MongoDB, FastAPI and Dash, with interpretable prioritisation signals and traceable evidence for human review.

[!\[\]\(f507db636256ac11a5525ef93ec6b8d7_img.jpg\) Data pipeline and review dashboard](#)

TEACHING, AWARDS & FUNDING

Calculus I Teaching Assistant, Universidad del Rosario

July 2024–July 2025

Led problem-solving sessions and supported first-year students.

Andrés Bello National Distinction

2024

One of 50 national-category recipients recognised by Colombia’s Ministry of Education for results in the 2023 Saber 11 national upper-secondary examination, used for university admissions.

TECHNICAL SKILLS & LANGUAGES

Programming Python, C, R, SQL.

Modelling & ML PyTorch, Gymnasium, SimPy, NumPy/SciPy, NetworkX, statsmodels.

Data & software SQLite, PostgreSQL, Polars, FastAPI, FAISS, Git, Linux, pytest, continuous integration, L^AT_EX.

Languages Spanish (native); English (C1; IELTS, 2023).